

PHYS 708 — Assignment 4 (Due date: 7 November 2025)

Problem

(1)

(a) The propagation of cosmic-ray primary species from a uniform distribution of sources in the Galactic disk follows

$$\nabla \cdot (D_p \nabla N_p) - 2h n v_p \sigma_p \delta(z) N_p = -S_p,$$

where $S_p(\mathbf{r}, E) = \mathfrak{R} Q_p(E) \delta(z)$ is the source term assumed to be extended up to the radial distance R . Using the solution of the above equation given on Slide 55 of Lecture 8, show that the spectrum in the Galaxy at very high energies follows a power law of the form $N_p(E) \propto E^{-(\gamma+a)}$ if the source spectrum and the diffusion coefficient follows $Q_p(E) \propto E^{-\gamma}$ and $D_p(E) \propto E^a$ respectively. [Hint: Refer to Slides 54-55 of Lecture 8].

(b) Under the same model given above, show that the spectrum of cosmic-ray secondaries at very high energies in the Galaxy follows $N_s(E) \propto E^{-(\gamma+2a)}$.

(c) Also show that the secondary-to-primary ratio follows an energy dependence of $\frac{N_s}{N_p} \propto E^a$. [Hint: Refer to Slide 58 of Lecture 8].

(2)

For a time-dependent propagation of cosmic-ray primaries from a nearby source, the transport equation is given by

$$\nabla \cdot (D_p \nabla N_p) - 2h n v_p \sigma_p \delta(z) N_p + \mathbb{Q}_p = \frac{\partial N_p}{\partial t},$$

where $\mathbb{Q}_p(\mathbf{r}, E, t) = Q_p(E) \delta(\mathbf{r}) \delta(t - t_0)$ is the source term injecting cosmic rays in a burst-like way. All other parameters have the same definition as in Question (1). Show that the spectrum at very high energies follows $N_p(E) \propto E^{-(\gamma+3a/2)}$. [Hint: Refer to Slide 70 of Lecture 8].